Wei-Jaw Lee

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Education

- **National Taiwan University** *2023/09 - Present*
  Ph.D. (direct-track from M.S. program), Graduate Institute of Communication Engineering Taipei, Taiwan
- **National Yang Ming Chiao Tung University** *2018/09 - 2023/01*
  B.S., Double Major in Biological Science & Technology and Electrical Engineering Hsinchu, Taiwan

Research Interest

- **Fine-grained evaluation and control:** Semantic alignment, concept-level evaluation, and feedback-driven generative modeling.
- **Controllable generation:** Multimodal conditioning and structured control for music and song generation.
- **Efficiency:** State-space models, data-efficient training, and computationally efficient music generation.

Publications

Conference Papers

[C.3] **Wei-Jaw Lee**, Hsuan-Yu Yeh, Ting-Yi Hu, Chih-Pin Tan, Fang-Duo Tsai, Yi-Hsuan Yang. "MPEcho: A Melody and Phoneme-Aware Generative Framework for Controllable Cover Song Generation," in *Proc. 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026.

[C.2] Fang-Chih Hsieh, **Wei-Jaw Lee**, Chun-Ping Wang, Hung-yi Lee, Hao-Wen Dong, Yi-Hsuan Yang. "Academic Text-to-Music Grand Challenge: Datasets, Baselines, and Evaluation Methods," in *IEEE International Conference on Multimedia and Expo (ICME), Grand Challenge*, 2026.

[C.1] Fang-Duo Tsai, Shih-Lun Wu, **Wei-Jaw Lee**, Sheng-Ping Yang, Bo-Rui Chen, Hao-Chung Cheng, Yi-Hsuan Yang. "MuseControlLite: Multifunctional Music Generation with Lightweight Conditioners," in *Proc. 42nd International Conference on Machine Learning (ICML)*, 2025.

Late-Breaking Demos

[L.2] Ping Wang, Fang-Duo Tsai, **Wei-Jaw Lee**, Yi-Hsuan Yang. "SheetSage-Pro: Bilingual Lyric-Aligned Lead-Sheet Transcription from Music Audio," *ISMIR Late-Breaking Demo*, 2026.

[L.1] **Wei-Jaw Lee**, Fang-Chih Hsieh, Xuanjun Chen, Fang-Duo Tsai, Yi-Hsuan Yang. "Exploring State-Space-Model Based Language Model in Music Generation," *ISMIR Late-Breaking Demo*, 2025.

Under Review

[U.2] **Wei-Jaw Lee**, Chun-Ping Wang, Xuanjun Chen, Yi-Hsuan Yang. "MQAScore: Fine-Grained Text-Music Alignment Evaluation via Large Audio Language Model," submitted to *IEEE ICASSP*, 2027.

[U.1] **Wei-Jaw Lee**, Fang-Chih Hsieh, Xuanjun Chen, Fang-Duo Tsai, Yi-Hsuan Yang. "Training-Efficient Text-to-Music Generation with State-Space Modeling," submitted to *IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*. arXiv:2601.14786.

Preprints

[P.1] Fang-Duo Tsai, Yi-An Lai, Fei-Yueh Chen, Hsueh-Wei Fu, **Wei-Jaw Lee**, Hao-Chung Cheng, Yi-Hsuan Yang. "MIDI-Informed Singing Accompaniment Generation in a Compositional Song Pipeline," arXiv:2602.22029, 2026.

Research Experience

- **Music and AI Lab, National Taiwan University** *2023/09 - Present*
  Advisor: Prof. Yi-Hsuan Yang Taipei, Taiwan
  - **Efficient Text-to-Music Generation with State-Space Models** [L.1], [U.1]
    - Led the development of a coarse-to-fine framework pairing an SSM-based language model for coarse codec tokens with a latent diffusion model for fine-grained tokens.
    - We demonstrate the training-efficiency of our proposed prefix SiMBA compared to transformer.
  - **Controllable Singing and Song Generation** [C.3], [C.1]
    - Led MPEcho, bridging singing voice synthesis and lyrics-to-song generation by introducing melody and phoneme-level control for cover song generation.

  - Contributed evaluation and analysis to MuseControlLite, a lightweight-conditioner approach for multifunctional music generation.
- **Evaluation and Benchmarking of Text-to-Music Systems** [U.2], [C.2]
  - Proposed MQAScore, a fine-grained text–music alignment metric built on a large audio language model.
  - Co-organized the ICME 2026 Academic Text-to-Music Grand Challenge; built the dataset and designed the concept coverage score.

Selected Coursework Projects

- **Deep Learning for Computer Vision: Final Project** *2024/12*
  VLLM, Corner Case Perception (task from ECCV 2024 Workshop Challenge) [GitHub]
  - Ranked 1st in the course competition as a team of four.
  - Designed dual adapters with LoRA fine-tuning to incorporate additional conditioning features, and a self-augmentation strategy that significantly improved the evaluation score.
- **Deep Learning for Music Analysis and Generation: Muse2bach** *2023/12*
  MIDI Style Transfer, Multi-Track Music Generation
  - Built a cover song style-transfer system combining lead sheet extraction (SheetSage) with multi-track generation (MMT); developed the multi-track generation module.
- **Computer Vision Practice with Deep Learning: VisProg Pro** *2023/12*
  Neuro-Symbolic AI, Programmatic Visual Reasoning
  - Extended VisProg (CVPR 2023 Best Paper) with an open-source LLM pipeline as an agentic function organizer and a depth-conditioned ControlNet module.

Honors and Awards

- **Top-Tier Doctoral Fellowship, College of Electrical Engineering and Computer Science** *2025 - 2026*
  National Taiwan University
- **Ministry of Education (MOE) Doctoral Scholarship** *2025 - 2026*
  Ministry of Education, Republic of China (Taiwan)

Experience

- **ML Research Intern** *2025/5 - 2026/10*
  Taiwan AILabs
  - Conducted research on controllable music generation and text–music evaluation, leading to two first-author papers: [C.3] (ISMIR 2026) and [U.2] (under review at ICASSP 2027).
- **Lead Teaching Assistant in Deep Learning for Music Analysis and Generation** *Fall 2023 – Fall 2025*
  National Taiwan University / Graduate Institute of Communication Engineering
  - Served as TA for three consecutive academic years.
  - Designed and graded homework assignments for students, covering topics such as pitch and instrument activities classification, MIDI generation.
  - Held weekly office hours and discussion sessions to support student learning and clarify course material.

Academic Service

- **Co-organizer, Academic Text-to-Music Grand Challenge** *2026*
  IEEE International Conference on Multimedia and Expo (ICME)
- **Reviewer**
  - **Journal:** TISMIR, IEEE TASLP, IEEE TCSVT
  - **Conference:** ISMIR 2025, ISMIR 2026
  - **Workshop:** NeurIPS 2025 AI for Music Workshop

Skills

- **Programming Languages:** Python, C, C++
- **Frameworks:** PyTorch, PyTorch Lightning, Hugging Face, Transformers, Diffusers
- **Tools:** Git, Weights & Biases, TensorBoard